NYSTATIN

Not absorbed from gut

Safe in infants

CHEMICAL NAME Polyene macrolide antifungal produced by *Streptomyces noursei* ($C_{47}H_{75}NO_{17}$)

CLASS Polyene antifungal (non-absorbed)

TRADE NAMES Mycostatin · Nystan · Nilstat

THERAPEUTIC USES / INDICATIONS

- Oral candidiasis (thrush) in infants, adults, denture wearers.
- Intestinal candidiasis.
- Cutaneous and vaginal candidiasis.
- Prophylaxis in immunocompromised patients and neonates (some units).

ADMINISTRATION

- **Oral suspension:** after meals, hold in mouth as long as possible then swallow; shake well.
- Infants: apply half the dose to each side of the mouth with dropper or swab.
- Remove dentures; clean them and apply nystatin to fitting surface.
- **Cream/ointment** twice daily; **pessaries** at night.

DOSAGES

- **Oral thrush (adult/child):** 100,000 units (1 mL) 4 times daily for 7 days; continue 48 h after cure.
- **Neonates:** 100,000 units 4 times daily.
- **Intestinal candidiasis:** 500,000 units 4 times daily.
- **Skin:** apply 2–4 times daily.

PHARMACOKINETICS

- Not absorbed from the GI tract, skin or mucosa.
- Excreted unchanged in faeces.

PHARMACODYNAMICS

- Binds **ergosterol** in fungal cell membrane → forms pores → leakage of cell contents → fungicidal.
- Active mainly against *Candida*.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, vomiting, diarrhoea (high doses).
- Oral irritation, bitter taste.
- Local irritation (topical).

ADVERSE EFFECTS (SERIOUS)

- Hypersensitivity, rash (rare); SJS (very rare).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to nystatin.

KEY INTERACTIONS

- No significant interactions (not absorbed).

NURSING MANAGEMENT

- Give after feeds/meals; avoid food/drink for 20–30 min after.
- Inspect mouth daily; document plaques.
- Check mother's nipples in breastfed infants; treat both.
- Sterilise bottles, teats and dummies.
- Assess for underlying causes (antibiotics, steroids, diabetes).

PATIENT EDUCATION

- Hold the liquid in the mouth before swallowing.
- Continue for 2 days after the mouth looks clear.
- Clean and soak dentures overnight.
- Rinse mouth after steroid inhalers.

KEY POINTS

- Polyene — not absorbed; safe in infants.
- Hold in mouth; continue 48 h after cure.
- Oral thrush: 1 mL QDS after feeds.
- Treat mother and baby together.